# Sheet 01: First Angle & Third Angle Projections

**BUREAU OF INDIAN STANDARDS (BIS SP: 46 - 2003 COMPLIANT)**

Title: Multiview Orthographic Projections & Standard Principles

**1. PROJECTION FUNDAMENTALS:**

- First Angle Projection: Object is placed in 1st Quadrant (Object lies between observer and plane).
  * Plan (Top View) lies BELOW Front Elevation.
  * Left View lies on the RIGHT side of Front View.
- Third Angle Projection: Object is in 3rd Quadrant (Plane lies between observer and object).
  * Plan lies ABOVE Front Elevation.
  * Left View lies on the LEFT side of Front View.

**2. STANDARD LINE TYPES & CONVENTIONS:**

- Continuous Thick Line (0.5mm): Visible outlines and edges.
- Continuous Thin Line (0.2mm): Dimension lines, extension lines, hatching lines.
- Dashed Medium Line (0.35mm): Hidden outlines and hidden edges.
- Long-dashed Dotted Thin (0.2mm): Centerlines, axes of symmetry, pitch circles.

**3. DIMENSIONING RULES:**

- Always place dimension figures above and clear of the dimension line (Aligned System).
- Extension lines must extend ~3mm beyond the dimension lines.
- Arrowheads must have length-to-depth ratio of 3:1.

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# Sheet 01 - Technical Exercises & Title Block

PRACTICE PROBLEMS & ASSIGNMENTS:

Problem 1: Given isometric block with dimensions 60 x 40 x 30 mm having a central step of 20 x 20 mm.
Draw the following views to scale 1:1 in First Angle Projection:
a) Front Elevation looking in direction of arrow X.
b) Top View (Plan).
c) Side Elevation from left.

**TITLE BLOCK SPECIFICATION (185 mm x 65 mm):**
[ NAME OF INSTITUTION / PUBLISHER: ALDAHR.ONLINE ]
[ TITLE: ENGINEERING DRAWING SHEET 01 ]
[ SCALE: 1:1 | PROJECTION: FIRST ANGLE | SHEET NO: 01 ]
[ CHECKED BY: VERIFIED CHIEF EXAMINER ]

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